

UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

Bao Nguyen • Fearghus Tyler • Yaqoub Rabiah • Fox Kang

Portland State University • MCECS • Department of Electrical & Computer Engineering

Sponsor: MME / Ichor Systems (Bridget Lannigan) | Advisors: Prof. Andrew Greenberg, Dr. Jonathan Bird (EE), Robert Paxton (ME)



The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for a new industrial-grade plastic shredder. The mechanical structure and cutting blades are delivered by a separate ME team.

Requirements

Must:

- Shred plastic to ~1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the \$1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.
May: read-only wireless monitoring, data logging, capacitive-touch safety.

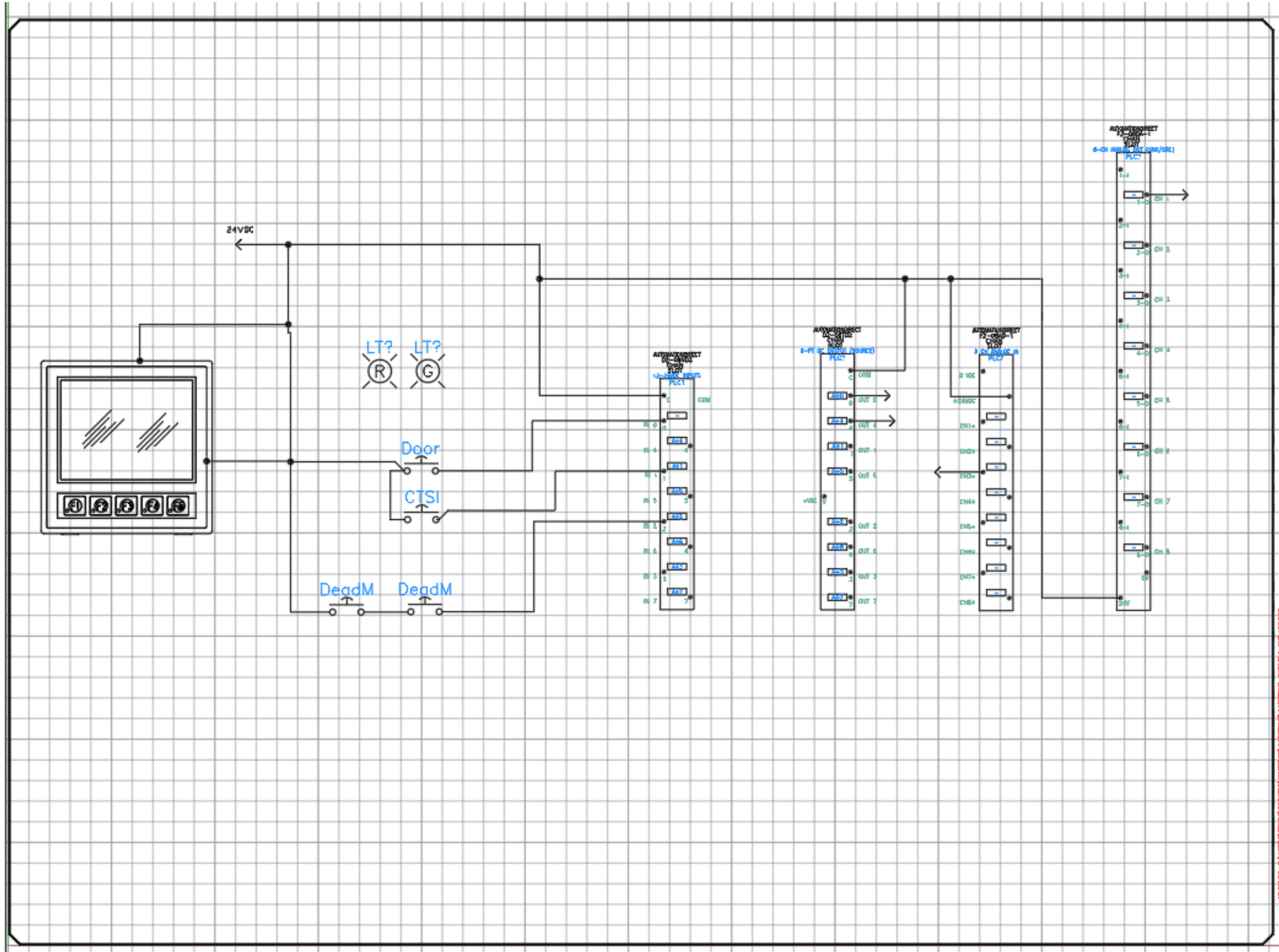
System Architecture



As-built operator control panel (left) and internal wiring (right): EA1-T4CL HMI, deadman / forward / reverse / E-stop pushbuttons, DirectLOGIC 205 rack, Huanyang VFD, Mean Well 24 VDC supply, SD-N35 contactor, and the fused terminal blocks.

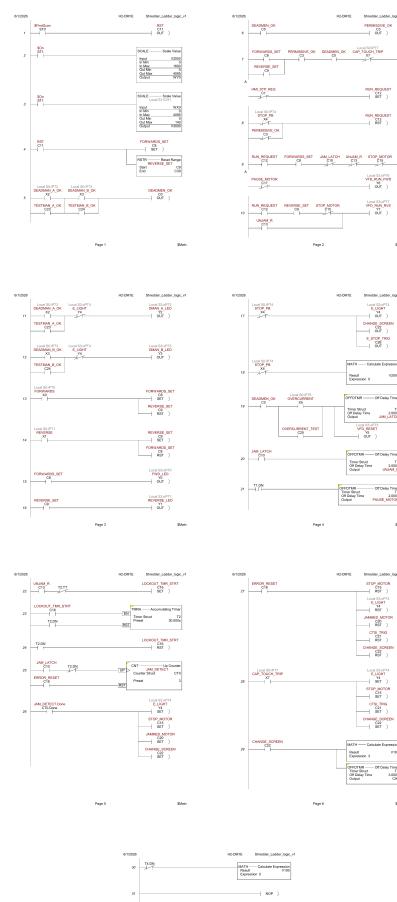
The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hardwired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 3 HP bench motor (GE 5KE182BC205B) over U/V/W via blue, white, and red 4 mm banana sockets.

PLC Control



DirectLOGIC 205 rack with H2-DM1E CPU and the five I/O modules.

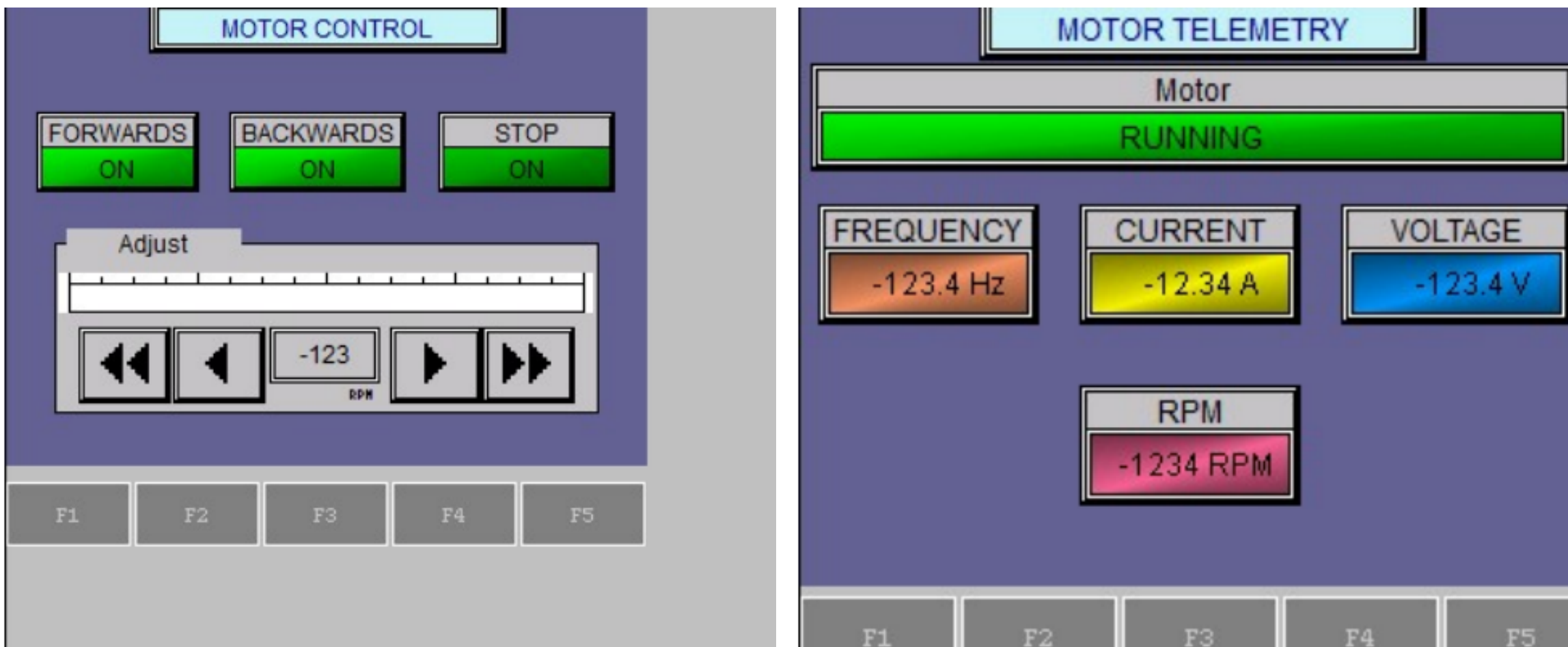
The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: D2-08ND3 (8 DC inputs for pushbuttons, E-stop monitor, VFD fault, capacitive touch), F2-08AD-1 (analog current input for VFD VO motor current), F2-04RTD (motor temperature, future), D2-08TR (8 relay outputs for VFD RUN FWD / RUN REV / RESET and panel LEDs), and F2-08DA-2 (0–10 V analog output to the VFD V1 speed reference). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.



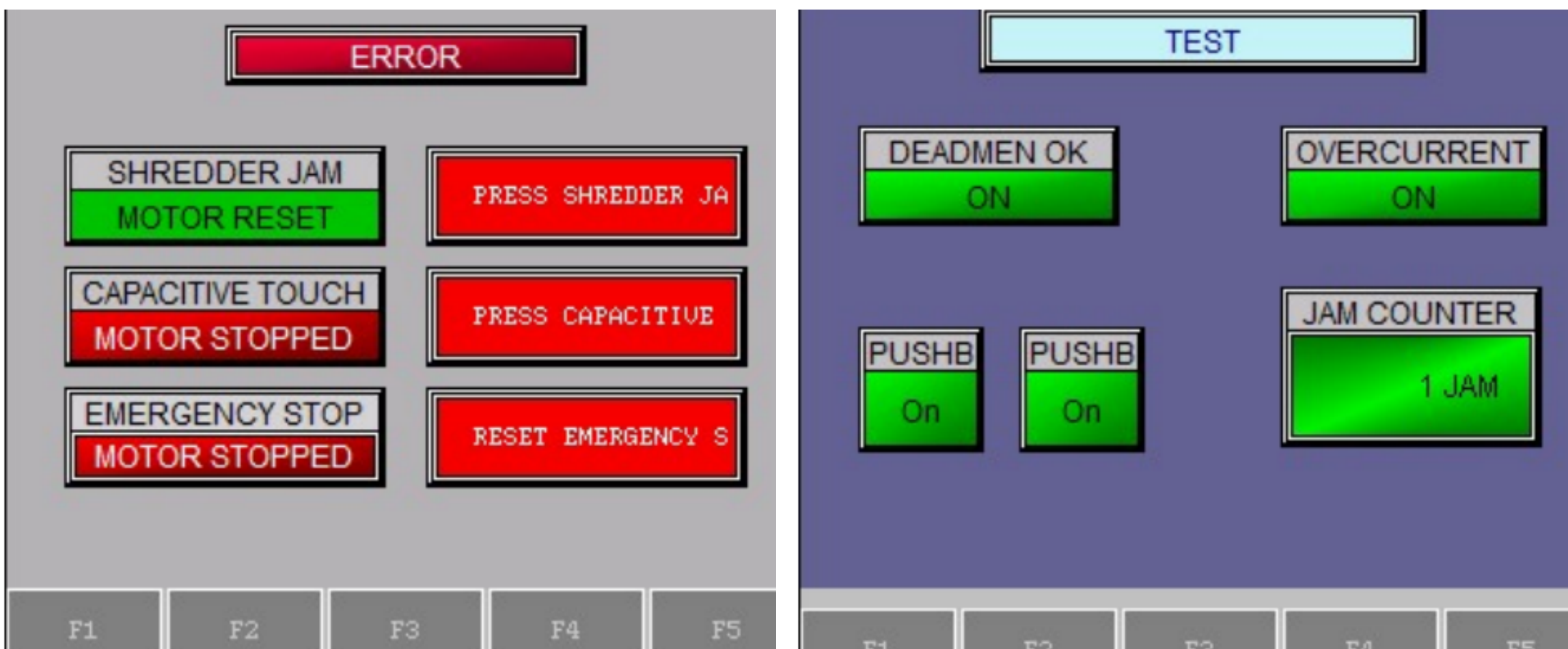
All 31 rungs (pages 1–7): init, deadman, RUN REQUEST gating, indicators, E-stop / jam detection, lockout, fault reset.

HMI

The operator interface is an AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running HMI_413.mgp, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags:



Motor Control • Motor Telemetry



Error / Alarm • Test / Commissioning

Safety & Compliance

- E-stop:** Category 0 stop. Cutler-Hammer E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3, with HMI soft mirrors C23 / C24).
- Overcurrent / jam:** VFD over-torque trip (PD124 = 150%, PD125 = 3 s) drives PLC X5 through the fault relay; ladder runs the 3-strike unjam routine.
- Enclosure:** Vorksmen SEB001 steel, manufacturer-rated **NEMA Type 12** (dust-tight, drip-tight).
- Compliance:** NFPA 70 (NEC), NFPA 79, NEMA 250 / ICS / MG 1, ISO 12100, ISO 13849-1 (preliminary Cat B–1 / PL b–c), ISO 13850, ISO 13851, OSHA 29 CFR 1910.147 (LOTO).

Results

- Panel built, wired, and bench-validated. All 31 ladder rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference path verified: HMI V2000 → SCALE → F2-08DA-2 CH1 → VFD V1.
- Jam-recovery verified: overcurrent on X5 latches C10 → 2 s reverse (T1) → coast-to-stop buffer (T3).
- Three-revision point list reviewed by Fearghus Tyler at each stage.

Future Work

- Production motor selection (3–5 HP) supersedes PD141--PD144.
- Mechanical integration / shred-load run pending ME team.
- Capacitive-touch module on X7: machine-level verification pending.
- Optional Modbus link to VFD for richer telemetry.

Acknowledgments

Thanks to **Bridget Lannigan** (MME / Ichor Systems) for sponsorship; to **Prof. Andrew Greenberg**, **Dr. Jonathan Bird**, and **Robert Paxton** for advising; and to the PSU Power Engineering Laboratory for bench access.